

Programmable On and Off Controller for 3-Phase Motor

DR R.V. DHEKALE

Automatic on/off controller for a 3-phase electric motor can be made with a programmable time switch. In this case a maximum of eight time durations can be programmed.

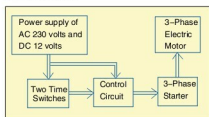


Fig. 1: Block diagram of the programmable on and off controller for a 3-phase electric motor

The system has two programmable time switches for setting the starting and stopping times of the motor and two control circuits, which are interfaced with the start and stop switches of the 3-phase motor's starter. Block diagram of the system is shown in Fig. 1.

Suppose, same clock times are set in both the time switches. So, if start

Test Points

Test point	Details
TP0	GND
TP1	+12V
TP2	+12V

time of, say, 8am is programmed for timer1 ON mode, then 8.01am will be programmed for timer1 OFF mode in start time switch. And, if stop time of, say, 9am is programmed for timer2 ON mode, then 9.01am will be programmed for timer2 OFF mode in stop time switch.

When time reaches 8am, the start time switch connects the primary of transformer X1 to 230V AC. Output of the power supply gets connected to reset pin 4 of IC1. R4 and C3 act as self triggering components.

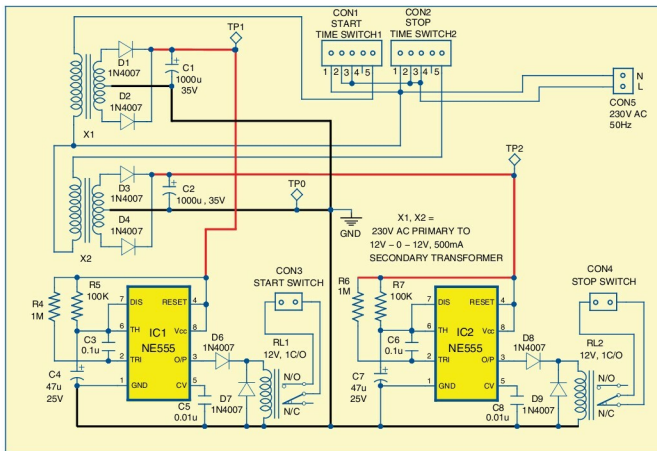


Fig. 2: Circuit diagram of the programmable on and off controller for a 3-phase electric motor

